

1 Show that the system of equations

$$\begin{array}{rcl} 14x - 4y + 6z & = & 5, \\ x + y + kz & = & 3, \\ -21x + 6y - 9z & = & 14, \end{array}$$

where k is a constant, does not have a unique solution and interpret this situation geometrically. [4]

[illegible]